



## Koneru Lakshmaiah Education Foundation

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SECTION : DBSE&DBD Y25 Section-8

## MYSQL STORED PROCEDURES



### MYSQL STORED PROCEDURES

*Examples and Practice Exercises for MySQL Workbench*

#### 1. Setup: Employees Table

```
CREATE DATABASE IF NOT EXISTS employeedb1;
```

```
USE employeedb1;
```

```
CREATE TABLE IF NOT EXISTS employees (
```

```
    emp_id INT PRIMARY KEY,
```

```
    emp_name VARCHAR(50),
```

```
    dept_id INT,
```

```
    salary DECIMAL(10,2)
```

```
);
```

```
INSERT INTO employees VALUES
```

```
(101,'Rahul',1,50000),
```

```
(102,'Priya',2,65000),
```

```
(103,'Anil',1,55000),
```

```
(104,'Sneha',3,70000),
```

```
(105,'Kiran',2,48000);
```

#### 2. Verify Data

Question: Display all employee records.

```
SELECT * FROM employees;
```

#### Procedure 1 – Display All Employees

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```

198 COMMIT;
199 END $$
200
201 DELIMITER ;
202
203 CALL TransferSalaryAmount(101,102,2000);
204
205 SELECT * FROM employees
206 WHERE emp_id IN (101,102);

```

Result Grid

| emp_id | emp_name | dept_id | salary   |
|--------|----------|---------|----------|
| 101    | Rahul    | 1       | 53000.00 |
| 102    | Priya    | 2       | 67000.00 |
| NULL   | NULL     | NULL    | NULL     |

Result 90 Result 91 employees 92 Result 93 Result 94 employees 95 Result 96 Result 97 Result 98 Result 99 Apply Revert

Output

Action Output

| #     | Time     | Action                                                          | Message                                                               | Duration / Fetch      |
|-------|----------|-----------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------|
| ✓ 453 | 12:44:41 | CALL AddBonus(@bonus)                                           | 0 row(s) affected                                                     | 0.000 sec             |
| ✓ 454 | 12:44:41 | SELECT @bonus AS UpdatedBonus LIMIT 0, 1000                     | 1 row(s) returned                                                     | 0.000 sec / 0.000 sec |
| ⚠ 455 | 12:44:41 | DROP PROCEDURE IF EXISTS TransferSalaryAmount                   | 0 row(s) affected, 1 warning(s): 1305 PROCEDURE employeeedb1.Trans... | 0.000 sec             |
| ✓ 456 | 12:44:41 | CREATE PROCEDURE TransferSalaryAmount( IN fromEmp INT, I...     | 0 row(s) affected                                                     | 0.016 sec             |
| ✓ 457 | 12:44:41 | CALL TransferSalaryAmount(101,102,2000)                         | 0 row(s) affected                                                     | 0.000 sec             |
| ✓ 458 | 12:44:41 | SELECT * FROM employees WHERE emp_id IN (101,102) LIMIT 0, 1000 | 2 row(s) returned                                                     | 0.000 sec / 0.000 sec |

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```
169 END $$
170
171 DELIMITER ;
172
173 • SET @bonus = 10000;
174
175 • CALL AddBonus(@bonus);
176
177 • SELECT @bonus AS UpdatedBonus;
```

Result Grid

| UpdatedBonus |
|--------------|
| 15000.00     |

employees 78 Result 79 Result 80 employees 81 Result 82 Result 83 employees 84 Result 85 Result 86 Result 87 Read Only x

Output

Action Output

| #     | Time     | Action                                                      | Message           | Duration / Fetch      |
|-------|----------|-------------------------------------------------------------|-------------------|-----------------------|
| ✓ 403 | 12:43:59 | SELECT @total AS TotalEmployees LIMIT 0, 1000               | 1 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ 404 | 12:43:59 | DROP PROCEDURE IF EXISTS AddBonus                           | 0 row(s) affected | 0.015 sec             |
| ✓ 405 | 12:43:59 | CREATE PROCEDURE AddBonus(INOUT bonus DECIMAL(10,2)) BEG... | 0 row(s) affected | 0.000 sec             |
| ✓ 406 | 12:43:59 | SET @bonus = 10000                                          | 0 row(s) affected | 0.000 sec             |
| ✓ 407 | 12:43:59 | CALL AddBonus(@bonus)                                       | 0 row(s) affected | 0.000 sec             |
| ✓ 408 | 12:43:59 | SELECT @bonus AS UpdatedBonus LIMIT 0, 1000                 | 1 row(s) returned | 0.000 sec / 0.000 sec |

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

```

153 SELECT COUNT(*) INTO total
154 FROM employees;
155 END $$
156
157 DELIMITER ;
158
159 • CALL EmployeeCount(@total);
160
161 • SELECT @total AS TotalEmployees;

```

Limit to 1000 rows

Result Grid

| TotalEmployees |
|----------------|
| 5              |

employees 57 Result 58 Result 59 employees 60 Result 61 Result 62 employees 63 Result 64 Result 65 Result 66 Read Only

Output

Action Output

| #     | Time     | Action                                                       | Message                                                              | Duration / Fetch      |
|-------|----------|--------------------------------------------------------------|----------------------------------------------------------------------|-----------------------|
| ✓ 311 | 12:42:33 | CREATE PROCEDURE SalaryStatus(IN eid INT) BEGIN DECLARE s... | 0 row(s) affected                                                    | 0.000 sec             |
| ✓ 312 | 12:42:33 | CALL SalaryStatus(102)                                       | 1 row(s) returned                                                    | 0.000 sec / 0.000 sec |
| ⚠ 313 | 12:42:33 | DROP PROCEDURE IF EXISTS EmployeeCount                       | 0 row(s) affected, 1 warning(s): 1305 PROCEDURE employeeedb1.Empl... | 0.000 sec             |
| ✓ 314 | 12:42:33 | CREATE PROCEDURE EmployeeCount(OUT total INT) BEGIN SEL...   | 0 row(s) affected                                                    | 0.016 sec             |
| ✓ 315 | 12:42:33 | CALL EmployeeCount(@total)                                   | 1 row(s) affected                                                    | 0.000 sec             |
| ✓ 316 | 12:42:33 | SELECT @total AS TotalEmployees LIMIT 0, 1000                | 1 row(s) returned                                                    | 0.000 sec / 0.000 sec |

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```

138 SELECT 'High Salary' AS Status;
139 ELSE
140 SELECT 'Low Salary' AS Status;
141 END IF;
142 END $$
143
144 DELIMITER ;
145
146 CALL SalaryStatus(102);

```

Result Grid

| Status      |
|-------------|
| High Salary |

employees 48 Result 49 Result 50 employees 51 Result 52 Result 53 employees 54 Result 55 Result 56 x Read Only

Output

Action Output

| #     | Time     | Action                                                       | Message                                                               | Duration / Fetch      |
|-------|----------|--------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------|
| ✓ 270 | 12:40:16 | DROP PROCEDURE IF EXISTS DepartmentAverageSalary             | 0 row(s) affected                                                     | 0.000 sec             |
| ✓ 271 | 12:40:16 | CREATE PROCEDURE DepartmentAverageSalary() BEGIN SELECT...   | 0 row(s) affected                                                     | 0.015 sec             |
| ✓ 272 | 12:40:16 | CALL DepartmentAverageSalary()                               | 3 row(s) returned                                                     | 0.000 sec / 0.000 sec |
| ⚠ 273 | 12:40:16 | DROP PROCEDURE IF EXISTS SalaryStatus                        | 0 row(s) affected, 1 warning(s): 1305 PROCEDURE employeeedb1.Salar... | 0.000 sec             |
| ✓ 274 | 12:40:16 | CREATE PROCEDURE SalaryStatus(IN eid INT) BEGIN DECLARE s... | 0 row(s) affected                                                     | 0.000 sec             |
| ✓ 275 | 12:40:16 | CALL SalaryStatus(102)                                       | 1 row(s) returned                                                     | 0.000 sec / 0.000 sec |

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```

116 FROM employees e
117 JOIN departments d
118 ON e.dept_id = d.dept_id
119 GROUP BY d.dept_id, d.dept_name;
120 END $$
121
122 DELIMITER ;
123
124 CALL DepartmentAverageSalary();

```

Result Grid

| dept_name | EmployeeCount | AverageSalary |
|-----------|---------------|---------------|
| CSE       | 2             | 55000.000000  |
| ECE       | 2             | 56500.000000  |
| EEE       | 1             | 70000.000000  |

employees 40 Result 41 Result 42 employees 43 Result 44 Result 45 employees 46 Result 47 x Read Only

Output

Action Output

| #   | Time     | Action                                                     | Message                                                              | Duration / Fetch      |
|-----|----------|------------------------------------------------------------|----------------------------------------------------------------------|-----------------------|
| 233 | 12:39:44 | CALL EmployeeDepartment()                                  | 5 row(s) returned                                                    | 0.000 sec / 0.000 sec |
| 234 | 12:39:44 | UPDATE employees SET salary = 55000 WHERE emp_id = 101     | 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0             | 0.016 sec             |
| 235 | 12:39:44 | SELECT * FROM employees WHERE emp_id = 101 LIMIT 0, 1000   | 1 row(s) returned                                                    | 0.000 sec / 0.000 sec |
| 236 | 12:39:44 | DROP PROCEDURE IF EXISTS DepartmentAverageSalary           | 0 row(s) affected, 1 warning(s): 1305 PROCEDURE employeeedb1.Depa... | 0.000 sec             |
| 237 | 12:39:44 | CREATE PROCEDURE DepartmentAverageSalary() BEGIN SELECT... | 0 row(s) affected                                                    | 0.015 sec             |
| 238 | 12:39:44 | CALL DepartmentAverageSalary()                             | 3 row(s) returned                                                    | 0.000 sec / 0.000 sec |

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```

92 SELECT e.emp_name, d.dept_name, e.salary
93 FROM employees e
94 JOIN departments d
95 ON e.dept_id = d.dept_id;
96 END $$
97
98 DELIMITER ;
99
100 • CALL EmployeeDepartment();

```

Result Grid

| emp_name | dept_name | salary   |
|----------|-----------|----------|
| Rahul    | CSE       | 75000.00 |
| Priya    | ECE       | 65000.00 |
| Anil     | CSE       | 55000.00 |
| Sneha    | EEE       | 70000.00 |
| Kiran    | ECE       | 48000.00 |

employees 27 Result 28 Result 29 employees 30 Result 31 Result 32 x Read Only

Output

Action Output

| #     | Time     | Action                                                               | Message                                                                       | Duration / Fetch      |
|-------|----------|----------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------|
| ✓ 168 | 12:36:42 | CALL AvgSalary()                                                     | 1 row(s) returned                                                             | 0.000 sec / 0.000 sec |
| ⚠ 169 | 12:36:42 | CREATE TABLE IF NOT EXISTS departments ( dept_id INT PRIMA...        | 0 row(s) affected, 1 warning(s): 1050 Table 'departments' already exists      | 0.000 sec             |
| ⚠ 170 | 12:36:42 | INSERT IGNORE INTO departments VALUES (1,'CSE'), (2,'ECE'), (3,'E... | 0 row(s) affected, 3 warning(s): 1062 Duplicate entry '1' for key 'departm... | 0.000 sec             |
| ✓ 171 | 12:36:42 | DROP PROCEDURE IF EXISTS EmployeeDepartment                          | 0 row(s) affected                                                             | 0.015 sec             |
| ✓ 172 | 12:36:42 | CREATE PROCEDURE EmployeeDepartment() BEGIN SELECT e.e...            | 0 row(s) affected                                                             | 0.000 sec             |
| ✓ 173 | 12:36:42 | CALL EmployeeDepartment()                                            | 5 row(s) returned                                                             | 0.000 sec / 0.000 sec |

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```
92 SELECT e.emp_name, d.dept_name, e.salary
93 FROM employees e
94 JOIN departments d
95 ON e.dept_id = d.dept_id;
96 END $$
97
98 DELIMITER ;
99
100 CALL EmployeeDepartment();
```

Result Grid

| emp_name | dept_name | salary   |
|----------|-----------|----------|
| Rahul    | CSE       | 75000.00 |
| Priya    | ECE       | 65000.00 |
| Anil     | CSE       | 55000.00 |
| Sneha    | EEE       | 70000.00 |
| Kiran    | ECE       | 48000.00 |

employees 27 Result 28 Result 29 employees 30 Result 31 Result 32 x Read Only

Output

Action Output

| #     | Time     | Action                                                               | Message                                                                       | Duration / Fetch      |
|-------|----------|----------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------|
| ✓ 168 | 12:36:42 | CALL AvgSalary()                                                     | 1 row(s) returned                                                             | 0.000 sec / 0.000 sec |
| ⚠ 169 | 12:36:42 | CREATE TABLE IF NOT EXISTS departments ( dept_id INT PRIMA...        | 0 row(s) affected, 1 warning(s): 1050 Table 'departments' already exists      | 0.000 sec             |
| ⚠ 170 | 12:36:42 | INSERT IGNORE INTO departments VALUES (1,'CSE'), (2,'ECE'), (3,'E... | 0 row(s) affected, 3 warning(s): 1062 Duplicate entry '1' for key 'departm... | 0.000 sec             |
| ✓ 171 | 12:36:42 | DROP PROCEDURE IF EXISTS EmployeeDepartment                          | 0 row(s) affected                                                             | 0.015 sec             |
| ✓ 172 | 12:36:42 | CREATE PROCEDURE EmployeeDepartment() BEGIN SELECT e.e...            | 0 row(s) affected                                                             | 0.000 sec             |
| ✓ 173 | 12:36:42 | CALL EmployeeDepartment()                                            | 5 row(s) returned                                                             | 0.000 sec / 0.000 sec |

MySQL Workbench

MYSQL x MYSQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```

77 CREATE TABLE IF NOT EXISTS departments (
78     dept_id INT PRIMARY KEY,
79     dept_name VARCHAR(50)
80 );
81
82 INSERT IGNORE INTO departments VALUES
83 (1, 'CSE'),
84 (2, 'ECE'),
85 (3, 'EEE');

```

Result Grid

| AverageSalary |
|---------------|
| 60600.000000  |

employees 16 Result 17 Result 18 employees 19 Result 20 x

Read Only

Output

Action Output

| #     | Time     | Action                                                              | Message                                                | Duration / Fetch      |
|-------|----------|---------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| ✓ 110 | 12:36:01 | SELECT * FROM employees WHERE emp_id = 101 LIMIT 0, 1000            | 1 row(s) returned                                      | 0.000 sec / 0.000 sec |
| ✓ 111 | 12:36:01 | DROP PROCEDURE IF EXISTS AvgSalary                                  | 0 row(s) affected                                      | 0.000 sec             |
| ✓ 112 | 12:36:01 | CREATE PROCEDURE AvgSalary() BEGIN SELECT AVG(salary) AS...         | 0 row(s) affected                                      | 0.000 sec             |
| ✓ 113 | 12:36:01 | CALL AvgSalary()                                                    | 1 row(s) returned                                      | 0.000 sec / 0.000 sec |
| ✓ 114 | 12:36:01 | CREATE TABLE IF NOT EXISTS departments ( dept_id INT PRIMA...       | 0 row(s) affected                                      | 0.047 sec             |
| ✓ 115 | 12:36:01 | INSERT IGNORE INTO departments VALUES (1,'CSE'), (2,'ECE'), (3,E... | 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 | 0.000 sec             |

MySQL Workbench

MySQL x MySQL x Local MySQL x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

```
68 • CREATE PROCEDURE AvgSalary()
69 BEGIN
70     SELECT AVG(salary) AS AverageSalary
71     FROM employees;
72 END $$
73
74 DELIMITER ;
75
76 • CALL AvgSalary();
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

| AverageSalary |
|---------------|
| 59600.000000  |

employees 11 Result 12 Result 13 employees 14 Result 15 x Read Only

Output

Action Output

| #    | Time     | Action                                                       | Message                                                             | Duration / Fetch      |
|------|----------|--------------------------------------------------------------|---------------------------------------------------------------------|-----------------------|
| ✓ 84 | 12:35:28 | CREATE PROCEDURE IncreaseSalary( IN eid INT, IN amount DE... | 0 row(s) affected                                                   | 0.015 sec             |
| ✓ 85 | 12:35:28 | CALL IncreaseSalary(101,5000)                                | 1 row(s) affected                                                   | 0.016 sec             |
| ✓ 86 | 12:35:28 | SELECT * FROM employees WHERE emp_id = 101 LIMIT 0, 1000     | 1 row(s) returned                                                   | 0.000 sec / 0.000 sec |
| ⚠ 87 | 12:35:28 | DROP PROCEDURE IF EXISTS AvgSalary                           | 0 row(s) affected, 1 warning(s): 1305 PROCEDURE employeeeb1.AvgS... | 0.000 sec             |
| ✓ 88 | 12:35:28 | CREATE PROCEDURE AvgSalary() BEGIN SELECT AVG(salary) AS...  | 0 row(s) affected                                                   | 0.000 sec             |
| ✓ 89 | 12:35:28 | CALL AvgSalary()                                             | 1 row(s) returned                                                   | 0.000 sec / 0.000 sec |

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```
48 • CREATE PROCEDURE IncreaseSalary(  
49     IN eid INT,  
50     IN amount DECIMAL(10,2)  
51 )  
52 BEGIN  
53     UPDATE employees  
54     SET salary = salary + amount  
55     WHERE emp_id = eid;  
56 END $$  
57  
58 DELIMITER ;  
59  
60 • CALL IncreaseSalary(101,5000);  
61  
62 • SELECT * FROM employees  
63 WHERE emp_id = 101;
```

Result Grid

| emp_id | emp_name | dept_id | salary   |
|--------|----------|---------|----------|
| 101    | Rahul    | 1       | 55000.00 |

Form Editor

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Navigator Query 1 x

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report
- Performance Schema

Administration

Information

No object selected

Object Info Session

Query 1

```

34
35 • CREATE PROCEDURE GetEmployeeById(IN eid INT)
36 BEGIN
37     SELECT * FROM employees
38     WHERE emp_id = eid;
39 END $$
40
41 DELIMITER ;
42
43 • CALL GetEmployeeById(102);

```

Result Grid

| emp_id | emp_name | dept_id | salary   |
|--------|----------|---------|----------|
| 102    | Priya    | 2       | 65000.00 |

employees 4 Result 5 Result 6 x

Output

Action Output

| #  | Time     | Action                                            | Message                                                 | Duration / Fetch      |
|----|----------|---------------------------------------------------|---------------------------------------------------------|-----------------------|
| 39 | 12:33:25 | DROP PROCEDURE IF EXISTS GetEmployees             | 0 row(s) affected                                       | 0.016 sec             |
| 40 | 12:33:25 | CREATE PROCEDURE GetEmployees() BEGIN SELECT...   | 0 row(s) affected                                       | 0.016 sec             |
| 41 | 12:33:25 | CALL GetEmployees()                               | 5 row(s) returned                                       | 0.000 sec / 0.000 sec |
| 42 | 12:33:25 | DROP PROCEDURE IF EXISTS GetEmployeeById          | 0 row(s) affected, 1 warning(s): 1305 PROCEDURE empl... | 0.000 sec             |
| 43 | 12:33:25 | CREATE PROCEDURE GetEmployeeById(IN eid INT) B... | 0 row(s) affected                                       | 0.016 sec             |
| 44 | 12:33:25 | CALL GetEmployeeById(102)                         | 1 row(s) returned                                       | 0.000 sec / 0.000 sec |

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Navigator Query 1 x

MANAGEMENT

- Server Status
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INSTANCE

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- Options File

PERFORMANCE

- Dashboard
- Performance Report
- Performance Schema

Administration

Information

No object selected

```
19 • DROP PROCEDURE IF EXISTS GetEmployees;
20
21 DELIMITER $$
22
23 • CREATE PROCEDURE GetEmployees()
24 BEGIN
25     SELECT * FROM employees;
26 END $$
27
28 DELIMITER ;
29
30 • CALL GetEmployees();
```

Result Grid

| emp_id | emp_name | dept_id | salary   |
|--------|----------|---------|----------|
| 101    | Rahul    | 1       | 50000.00 |
| 102    | Priya    | 2       | 65000.00 |
| 103    | Anil     | 1       | 55000.00 |
| 104    | Sneha    | 3       | 70000.00 |
| 105    | Kiran    | 2       | 48000.00 |

Object Info Session employees 2 Result 3 x

Read Only

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Navigator Query 1 x

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report
- Performance Schema

Administration

Information

No object selected

Query 1

```
1 • CREATE DATABASE IF NOT EXISTS employeeedb1;
2 • USE employeeedb1;
3 • USE employeeedb1;
4 • CREATE TABLE IF NOT EXISTS employees (
5     emp_id INT PRIMARY KEY,
6     emp_name VARCHAR(50),
7     dept_id INT,
8     salary DECIMAL(10,2)
9 );
10
11 • INSERT INTO employees VALUES
12 (101, 'Rahul', 1, 50000.00);
13 (102, 'Priya', 2, 65000.00);
14 (103, 'Anil', 1, 55000.00);
15 (104, 'Sneha', 3, 70000.00);
16 (105, 'Kiran', 2, 48000.00);
```

Result Grid

| emp_id | emp_name | dept_id | salary   |
|--------|----------|---------|----------|
| 101    | Rahul    | 1       | 50000.00 |
| 102    | Priya    | 2       | 65000.00 |
| 103    | Anil     | 1       | 55000.00 |
| 104    | Sneha    | 3       | 70000.00 |
| 105    | Kiran    | 2       | 48000.00 |
| * NULL | NULL     | NULL    | NULL     |

Insert new row

Object Info Session employees 1 x

Apply Revert